Arithmetic Sequences and Series

ANSWER KEY

Section 1: General Term of an Arithmetic Sequence

1 State the formula for the n-th term a_n of an arithmetic sequence with first term a_1 and common difference d.

a. $a_n = a_1 + (n-1)d$ ✓

b. $a_n = a_1 \cdot d^{n-1}$

c. $a_n = a_1 + nd$

d. $a_n = \frac{a_1 + a_n}{2}$

In an arithmetic sequence, each term increases by d, so the n-th term is $a_n = a_1 + (n-1)d$.

2 Find the 20th term of the arithmetic sequence $4, 7, 10, 13, \ldots$

a. 61 ✓

b. 58

c. 64

d. 60

$a_1 = 4, d = 3$. $a_{20} = 4 + (20-1)(3) = 4 + 19(3) = 4 + 57 = 61$.

3 In an arithmetic sequence, $a_3 = 11$ and $a_8 = 31$. Find the first term a_1 and common difference d.

a. $a_1 = 3, d = 4$ ✓

b. $a_1 = 4, d = 3$

c. $a_1 = 2, d = 5$

d. $a_1 = 1, d = 4$

$a_8 - a_3 = 5d = 31 - 11 = 20 \Rightarrow d = 4$. $a_1 = a_3 - 2d = 11 - 2(4) = 3$.

Section 2: Sum of an Arithmetic Series

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4 State the two standard formulas for the sum S_n of the first n terms of an arithmetic series.

a. $S_n = \frac{n}{2}(a_1 + a_n)$ **and** $S_n = \frac{n}{2}[2a_1 + (n-1)d]$ ✓

b. $S_n = \frac{a_1(1-r^n)}{1-r}$

c. $S_n = n(a_1 + a_n)$

d. $S_n = \frac{a_1 + a_n}{2}$

Gauss's pairing formula gives $S_n = \frac{n}{2}(a_1 + a_n) = \frac{n}{2}[2a_1 + (n-1)d]$.

5 Find the sum of the first 30 terms of the arithmetic series $2 + 5 + 8 + 11 + \ldots$

a. 1365 ✓

b. 1350

c. 2730

d. 1410

$n = 30, a_1 = 2, d = 3$. $S_{30} = \frac{30}{2}[2(2) + (29)(3)] = 15[4 + 87] = 15(91) = 1365$.

6 Find the sum of the positive integers from 1 to 100.

a. 5050 ✓

b. 5000

c. 5100

d. 10100

$S_{100} = \frac{100(101)}{2} = 50 \times 101 = 5050$.

Section 3: Sigma Notation and Number of Terms

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7 Evaluate $\sum_{k=1}^{25}(4k-1)$.

a. 1275 ✓

b. 1300

c. 1250

d. 2550

First term $a_1 = 4(1) - 1 = 3$. Last term $a_{25} = 4(25) - 1 = 99$. Number of terms $n = 25$.
$S_{25} = \frac{25}{2}(3 + 99) = \frac{25}{2}(102) = 25(51) = 1275$.

8 How many terms are in the finite arithmetic sequence $7, 11, 15, \ldots, 87$?

a. 21 ✓

b. 20

c. 22

d. 19

$a_n = a_1 + (n-1)d \Rightarrow 87 = 7 + (n-1)(4) \Rightarrow 80 = 4(n-1) \Rightarrow 20 = n - 1 \Rightarrow n = 21$.

9 Evaluate $\sum_{k=5}^{20} 3k$.

a. 600 ✓

b. 570

c. 630

d. 300

Number of terms $n = 20 - 5 + 1 = 16$. $a_1 = 3(5) = 15$, $a_{16} = 3(20) = 60$.
$S = \frac{16}{2}(15 + 60) = 8(75) = 600$.

Section 4: Word Problems and Real-World Applications

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10 An auditorium has 20 rows of seats. The first row has 15 seats, and each subsequent row has 2 more seats than the row before it. How many total seats are in the auditorium?

a. 680 ✓

b. 660

c. 700

d. 580

$a_1 = 15, d = 2, n = 20$. $S_{20} = \frac{20}{2}[2(15) + (19)(2)] = 10[30 + 38] = 10(68) = 680$.

11 A runner trains by running 3 miles on day 1, and increases her daily distance by 0.5 miles each day. How many total miles will she have run in 14 days?

a. 87.5 miles ✓

b. 91 miles

c. 84 miles

d. 95.5 miles

$S_{14} = \frac{14}{2}[2(3) + (13)(0.5)] = 7[6 + 6.5] = 7(12.5) = 87.5$ miles.

12 A clock chimes the number of hours on the hour (1 chime at 1:00, 2 chimes at 2:00, ..., 12 chimes at 12:00). How many total chimes does it make in a 12-hour period?

a. 78 ✓

b. 72

c. 84

d. 66

Sum $= \frac{12(13)}{2} = 6 \times 13 = 78$.

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